IMPERIAL COLLEGE LONDON

EE4-17
EE9-AO4

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc and EEE PART IV: MEng and ACGI

HIGH PERFORMANCE ANALOGUE ELECTRONICS

Tuesday, 19 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : A.S. Holmes

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The Questions

1.

(a) The RF signal received by a heterodyne receiver is centred at 10MHz. If the image signal is at 15MHz, what would be the frequency of the local oscillator?

[3]

(b) A prefilter for the receiver in (a) is designed as a second order low pass function with a cut-off frequency of 12MHz, a DC gain of 1, and a quality factor of 2. Write the transfer function of the filter.

[3]

(c) What would be the attenuation of the image signal after the filter in (b)?

[3]

(d) In a receiver, a bandpass filter is used to remove image signals located in a 12.3MHz to 12.8MHz band. What is the bandwidth of the RF signal of interest, if the local oscillator frequency is 10.7MHz?

[3]

(e) In a receiver, what are the advantages and disadvantages of having a high gain in the front-end low noise amplifier (LNA)?

[3]

(f) The transfer function of a circuit is:

$$\frac{5.685\cdot10^{15}}{s^2+1.066\cdot10^{8}s+\cdot5.685\cdot10^{15}}$$

What type of circuit is it? What would you use this type of circuit for in a receiver architecture?

[3]

(g) Figure 1.1 shows a LC ladder with the same characteristics as the circuit in (f). Find an equivalent implementation of the ladder using only integrators and amplifiers.

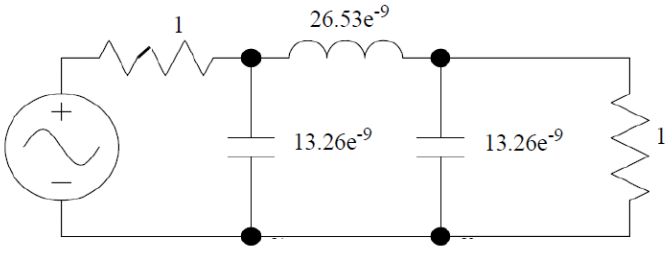


Figure 1.1

[2]

2.

(a) Explain why transforming a circuit topology into a differential configuration can be beneficial when the linear range needs to be extended.

[2]

(b) Explain in your own very simple words (i.e. without copying from the lecture notes), why someone might consider to use degeneration in a differential pair.

[3]

(c) In a differential pair with degeneration and resistors as load and biasing, draw all the different noise sources and briefly explain which noises they correspond to.

[3]

(d) Without carrying out any analysis, explain which would be the dominant noise sources in a differential pair without degeneration resistors.

[3]

(e) What would be the advantages and disadvantages of removing the degeneration in the differential pair.

[3]

(f) A system is formed by a cascade of three blocks. The first and third blocks have an input referred white noise power of $1(\mu V)^2$, integrated in a 0 to 100Hz bandwidth. The second block has an output referred white noise density of $4(\mu V)^2/Hz$. The power gain of the first block is 1. The power gain of the second block is 0.5. The power gain of the third block is 10. The input noise power from the source is $1(\mu V)^2$ in a 10kHz to 20kHz bandwidth. What is the minimum input signal in a 1kHz to 20kHz bandwidth that the system can process whilst being above the noise floor at the output?

[3]

(g) As a designer, what do you think could have been done differently to improve the noise performance of the previous system?

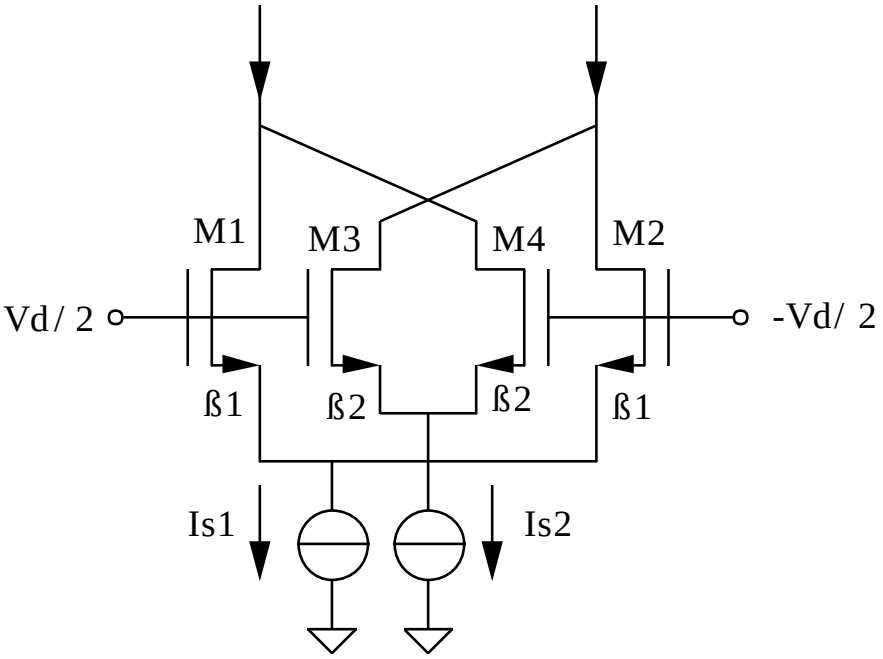
[3]

3.

(a) Explain qualitatively and in your own words how by choosing different sizes in the two input transistors in a differential pair, and combining two of those differential pairs, you can extend the linearity of the conventional differential pair. You can use a graph to support your explanation.

[4]

(b) In the circuit below, why do the two differential pairs have different sized transistors? (Explain with words)



[4]

(c) Derive an expression for the transconductance of the circuit in (b) without copying the derivation from the lecture notes.

[4]

(d) Explain with words what would happen if the transistors in the individual differential pairs were mismatched.

[4]

(e) You designed the circuit in (b), sent it for fabrication, and during testing your notice that when inputting a sinusoidal signal, your output deviates a bit from the sinusoidal shape. What do you think is happening?

[4]

4.

(a) A designer starts designing the transconductor shown in Figure 4.1 in Cadence. The amplifier has a gain over 1000. When looking at the output of the transient simulation she observes something is not right. What does she see?

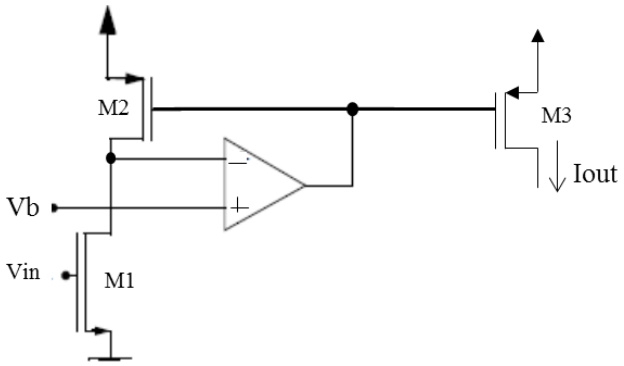


Figure 4.1

[4]

(b) Based on what she sees, she looks at the circuit and decides to swap the connections to the amplifier, as shown below (Note: Positive and negative inputs swapped). Why does she do that?

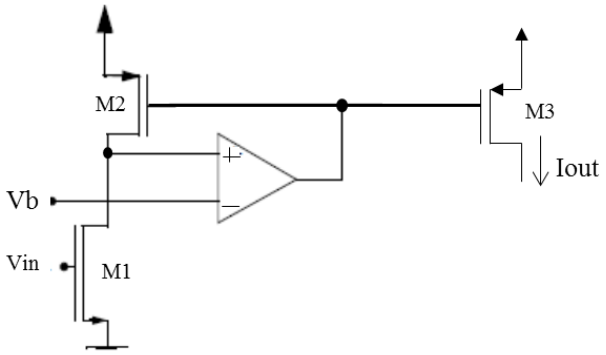


Figure 4.2

[4]

(c) The circuit now works better (i.e. the effect she was seeing in (a) is not there anymore), but she observes the output is not very linear. She then finds the reason for that. Which transistor is likely to be biased in the wrong operating region, and what should the operating regions of the different transistors be to maximize linearity?

[4]

(d) Derive an expresion for the output current of the circuit in (b)

[4]

(e) Derive an expression for the transconductance of the circuit in (b).

[4]